

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Wrote and tested systems-level infrastructure code in **Go**, **Python**, and **C++** for scheduling, resource allocation, and fault-tolerant placement across a **large-scale** multi-cloud (**AWS**, **GCP**, **CoreWeave**) **compute** fleet under capacity constraints.
- Participated in design reviews with peers and stakeholders to decide amongst available technologies for GPU fleet orchestration, balancing training/inference workload demand against **hardware** efficiency.
- Reviewed code from other developers for style, testability, and efficiency, and **triaged** production issues, **debugging**/tracking/resolving defects by analyzing impact on **hardware**, **network**, and service operations.
- Built observability and autoscaling for inference serving (Prometheus/Grafana), using **network** and **storage** telemetry to diagnose fleet-wide performance regressions.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Operated multi-cloud **GPU** infrastructure with **Docker/Kubernetes**-based scheduling, writing **Go** and **Python** tooling for fault-tolerant placement under node loss and provider hiccups.
- Cut p99 serving latency **420ms to 140ms** across 50K+ daily requests via a systems-level async redesign and performance tuning, documenting design decisions for the team.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and shipped **25+ C++/Java distributed systems** serving **1M+ users** at **99.9% availability** over 4 years on **AWS** (EC2, RDS, SQS), leading **design and code reviews** and driving **software design and architecture** decisions across service boundaries, APIs, and concurrency relied on by other engineers.
- Tested, maintained, and launched production services; standardized API contracts and failure-handling across 25+ services, reducing escalations **60%** via consistent retry, timeout, and testing practices.
- Triaged and **debugged**/tracked/resolved production issues across application, **network**, and service layers using logs, metrics, and root-cause analysis, feeding fixes back into platform standards via code review.
- Redesigned nightly **SQL** data aggregation blocking downstream settlement, cutting runtime **8 minutes to 90 seconds** via indexed views and batch partitioning, eliminating row-level lock contention at scale.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Built a fault-tolerant distributed **KV store** in **C++** using **Raft consensus**: leader election, replicated logs, **concurrent writes**, **linearizable reads**, and snapshot/log compaction with recovery under node failures.
- Built a companion **VPC control-plane** simulator modeling distributed **network** state and route propagation across nodes, exploring consistency and **control-plane** convergence under topology changes.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent system automating production incident diagnosis via tool-use chains, applying object-oriented design to route alerts and synthesize root cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Go, Python, C++, Java, SQL, HTML5, CSS, React

Fundamentals: Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks

Infrastructure: Distributed Systems, Large-Scale Compute, GPU/Hardware Architecture, Storage, Networking

Cloud & DevOps: AWS, GCP, CoreWeave, Docker, Kubernetes, CI/CD, Terraform

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer